

In Vivo Ultrasound Imaging of Macrophages Using Acoustic Vaporization of Internalized Superheated Nanodroplets

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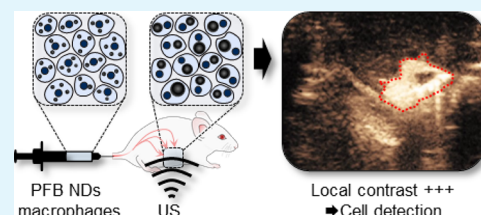
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ABSTRACT: Activating patients' immune cells, either by reengineering them or treating them with bioactive molecules, has been a breakthrough in the field of immunotherapy and has revolutionized treatment, especially against cancer. As immune cells naturally home to tumors or injured tissues, labeling such cells holds promise for non-invasive tracking and biologic manipulation. Our study demonstrates that macrophages loaded with extremely low boiling point perfluorocarbon nanodroplets not only survive ultrasound-induced phase change but also maintain their phagocytic function. Unlike observations made when using higher boiling point perfluorocarbon nanodroplets, our results show that phase change occurs intracellularly at a low mechanical index using a clinical scanner operating within the energy limit set by the Food and Drug Administration (FDA). After nanodroplet-loaded macrophages were given intravenously to nude rats, they were invisible in the liver when imaged at a very low mechanical index using a clinical ultrasound scanner. They became visible when power was increased but still within the FDA limits up to 8 h after administration. The acoustic labeling and in vivo detection of macrophages using a clinical ultrasound scanner represent a paradigm shift in the field of cell tracking and pave the way for potential therapeutic strategies in the clinical setting.

KEYWORDS: cell-tracking, intracellular acoustic droplet vaporization, perfluorobutane nanodroplets, ultrasound



1. INTRODUCTION

In oncology, in vivo immune cell tracking has the potential to provide valuable insight into the tumor microenvironment, cellular trafficking, safety, and early response to treatment.¹ Direct methods for cell labeling and non-invasive in vivo tracking include the use of radioisotopes for single-photon emission computed tomography or positron emission tomography,² bioluminescent or fluorescent probes for optical imaging, perfluorocarbon (PFC) nanoemulsions for ¹⁹F magnetic resonance imaging (MRI), or PFC gas microbubbles (MBs) for ultrasound (US) imaging.^{3,4} Monitoring T-cell homing to tumors by labeling them and imaging with PET has shown promise in preclinical models but has not yet been translated to patients.⁵ Recent reports have shown that labeling T-cells,^{6–9} natural killer^{10,11} (NK), and dendritic cells^{12–14} ex vivo with PFC NDs allowed their visualization and quantification in vivo using ¹⁹F MRI in mice and is currently in early stage clinical trials.^{15,16} Importantly, the presence of PFC nanodroplets (NDs) in NK cells did not interfere with their function in vitro and in mice.^{11,17} Further, when ¹⁹F MRI quantification showed an increase in CAR T-cell homing and persistence in glioblastoma xenografts, it correlated with tumor regression.⁷ In another study, PFC-labeled macrophages were used as a means to assess local inflammation in a mouse model of inflammatory bowel disease by quantifying cell accumulation using ¹⁹F MRI.¹⁸

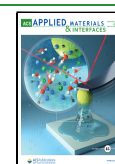
Labeling cells with acoustically responsive reporters to allow their detection in vivo by US presents advantages over PET and MRI: US is extremely sensitive to MBs¹⁹ and is capable of detecting a single acoustically labeled cell in vivo.^{20,21} In addition, US is the only imaging modality that not only detects but can also cause local mechanical effects when interacting with reporters such as PFC-filled NDs, nanobubbles (NBs), or MBs in vivo.^{22–26} For example, loading DNA on the shell of internalized MBs allowed the transfection of cells in vitro²⁰ and in vivo²⁶ using a planar wave single element transducer (Sonigene sonoprotector) operating at 1 or 2 W/cm². This result indicates that acoustically visible and responsive materials can be used as theranostic agents to not only image tissues or cells but also allow for drug or gene delivery to deep tissues with spatial and temporal control.

MBs are the spearhead of acoustically responsive reporters due to their established safety and current use in the clinic as diagnostic agents. Although the acoustic labeling of neuroprogenitor cells for in vivo imaging using MBs has been reported, the ability to load and image internalized low boiling

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point NDs has not been explored and could provide several advantages:²⁰ (i) because of their ability to vaporize into echogenic MBs using clinical scanners, stable superheated PFC NDs can act as MB precursors; (ii) similar to MBs whose longevity increased from a few minutes to a few days when internalized by neuroprogenitor cells,²⁰ we expect that NDs of low boiling point PFCs will also experience greater longevity when internalized. Greater longevity relative to free NDs in circulation would be mostly due to the protection provided by the lipid bilayers of endosomes and cell membranes when cells are exposed to US or when the NDs traverse lung capillaries. (iii) Finally, because more NDs can be loaded intracellularly than MBs, the increase in shell surface area is expected to increase drug load. Beyond the scope of this work, we foresee that a much lesser cell count would be needed for detection compared to the less sensitive MRI method. However, unlike ¹⁹F MRI, the acoustic labeling of immune cells and US tracking will likely be non-quantitative but will rather provide qualitative insight into immune cell trafficking and pave the way for US-mediated immune stimulation by preloading the NDs with immunomodulating drugs.

Other than internalized MBs in neuroprogenitor cells, there is currently no other platform for cell labeling and in vivo tracking that combines single-cell sensitivity while maintaining cell viability and biological function.^{20,26} Dodecafluoropentane (DDFP) or doxorubicin-loaded DDFP micron-sized droplets (boiling point, BP = 28 °C) internalized in macrophages could be vaporized in vitro.^{27,28} However, acoustic droplet vaporization (ADV) required the use of US energies higher than an FDA limit of 1.9 mechanical index (MI), limiting the use of clinical US systems. Xu *et al.* also used cell killing and fragmentation by US by pre-loading macrophages with both DDFP NDs and doxorubicin that they then released locally in tumor-bearing mice using high intensity focused ultrasound (HIFU) at 1 MHz, a concept they named “ultrasound-triggered cell bomb”.²⁹ Pertinent to this study, our group reported the direct manufacturing of highly concentrated perfluorobutane (PFB) NDs that remain stable in suspension during formulation and antibody conjugation and for at least 2 h in vitro at 37 °C without vaporizing into MBs until US was applied at ≥0.4 MI, well below the FDA limit.^{30,31} We aimed to test our hypotheses that the ADV of PFB NDs preloaded into macrophages occurs in vitro, will keep cells alive and functional, and will occur in vivo using a clinical scanner to ease clinical translation.

Intracellular labeling of human THP-1 derived macrophages was done in vitro using superheated PFB NDs, and droplet vaporization was performed using a clinical US scanner (Siemens ACUSON Sequoia). Cell survival post ND internalization and intracellular vaporization was assessed using viability dyes, and the preservation of phagocytic function was confirmed using a phagocytosis assay. As proof that intracellular PFB NDs can vaporize in vivo, PFB ND-loaded human macrophages were infused into nude rats and allowed to sequester in the liver and ADV performed up to 8 h after infusion using built-in MB destruction pulses that are used clinically to eliminate MBs from the field of view to observe the refill rates.

2. MATERIALS AND METHODS

2.1. Materials. 1,2-Distearoyl-*sn*-glycero-3-phosphocholine (DSPC) and 1,2-distearoyl-*sn*-glycero-3-phosphoethanolamine-*N*-methoxy(polyethyleneglycol)-2000 (DSPE-PEG2K) were purchased

from NOF America (NY, USA). 1,2-Stearoyl-3-trimethylammonium-propane (DSTAP) was purchased from Avanti Polar lipids (AL, USA). DSPE-PEG2K-Cy5 was purchased from Biohempeg (MA, USA). Fetal bovine serum (FBS), RPMI-1640, and THP-1 cell line were purchased from American Type Culture Collection (VA, USA). 2-Mercaptoethanol, Dulbecco's phosphate buffered-saline (DPBS), Calcein-AM, live/dead viability/cytotoxicity kit, pHrodo Deep Red *Escherichia coli* BioParticles Conjugate, and sodium hyaluronate (Lifecore biomedical LLC) were purchased from Thermo Fisher Scientific (MA, USA). Perfluorobutane was purchased from SynQuest Labs (AL, USA). AOP1 was purchased from Nexelcom Bioscience (MA, USA). Phorbol 12-myristate 13-acetate (PMA) and deoxyribonuclease I (DNase) were purchased Sigma-Aldrich (MO, USA). Cell Counting Kit 8 (WST-8/CCK8) was purchased from Abcam. All reagents were used without further purification.

2.2. Nanodroplet Formulation and Characterization. NDs were prepared following a direct emulsification procedure using phospholipid films made of DSPC, DSPE-PEG2K, and DSTAP at 80:10:10, 70:10:20, and 60:10:30 molar ratios, respectively. Briefly, lipid films were hydrated in PBS 1×/propylene glycol/glycerol (80:15:5 v/v/v, 2 mL at 3.5 mg/mL) and sonicated at 70 °C until clear. After being cooled down to −12 °C using a salt and ice water bath, the phospholipid suspension was added to 200 μL of condensed PFB in a −12 °C salt and ice water bath. The mixture was emulsified using a probe sonicator (Branson, 20%, 10 s). The NDs were collected by centrifuging the emulsion at 200g for 1 min and discarding the supernatant. NDs were then stored in a glass vial at 4 °C and were used within 2 weeks. The PFB content in the emulsion was quantified by gas chromatography (GC)/mass spectroscopy (MS) as follows: A standard curve was first generated by adding from 1 to 250 μL of PFB gas in air-filled 25 mL Teflon-sealed vials to produce PFB:air ratios from 0.004 to 1.000% v/v. Samples (10 μL) from each vial were injected into the port of an Agilent Technologies 7820A-GC associated with an Agilent 5975-MSD spectrometer. Ten percent of the injected volume flowed through a Restek Rxi-5Sik MS column (30 m × 0.25 mm × 0.25 μm) driven by helium heated to 40 °C and flowing at 0.5 mL/min. PFB eluted at 1.1 min and was identified by electron impact ionization (*m/z* = 69 and 119). The area under the curve in arbitrary units done in triplicate was plotted against the PFB concentration and the slope of the line calculated. The PFB content of the emulsion was measured by placing 50 μL in a 1 mL syringe and heating to 100 °C in a water bath with the plunger pointing up. The gas (10 μL) generated was added to a 25 mL air-filled vial and 10 μL of the mixture analyzed as above. The area under the curve generated was converted to the PFB/air concentration using the slope of the standard regression line.

To produce fluorescently labeled NDs for some of the subsequent experiments, either 2% of the DSPE-PEG2K was substituted with the fluorescently labeled DSPE-PEG2K-Cy5 or 10 μL of 1 mM DiD solution was added to the lipid/PFB mixture prior to sonication.

NDs used in subsequent experiments were characterized prior to each use by assessing size, count, and zeta potential using a ZetaView (Particle Metrix) nanoparticle tracking analyzer (NTA). When not specified, the term NDs or PFB NDs refer to PFB NDs containing 30% DSTAP.

ND stability in tissue culture media was monitored using dynamic light scattering (DLS) by assessing intensity-weighted size distributions, derived count rates, and correlation curves as a function of incubation time at 37 °C. Briefly, 50 μL NDs were added to either 350 μL of milliQ water or 350 μL of RPMI-1640 and heated at 37 °C for 15, 30, and 60 min. DLS measurements were performed with a Zetasizer ZS nano-sizing system (Malvern Nano ZS) at baseline in milliQ water and media and after 15, 30, and 60 min of incubation at 37 °C in media. Samples were monitored for ND loss and MB formation due to thermal instability³⁰ as well as ND agglomeration due to the formation of protein corona in the ND surface.³²

2.3. THP-1 Derived Macrophage Preparation. Human THP-1 monocytes were differentiated into macrophages using reported procedures.^{33–35} Briefly, 10⁶ THP-1 cells were placed in six-well plates and cultured for 72 h in 2 mL of the base medium (RPMI 1640

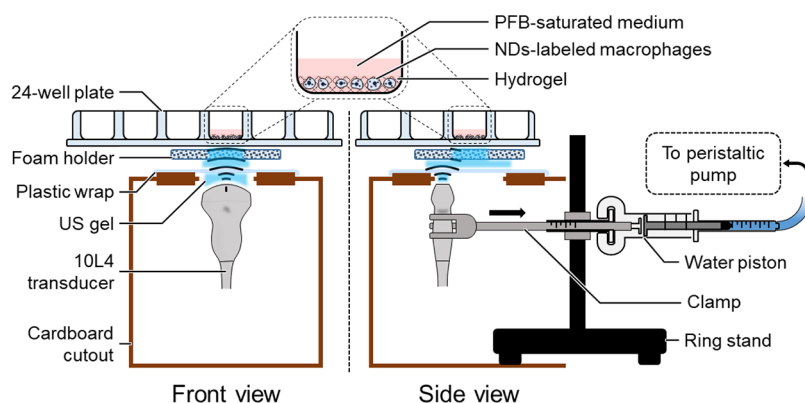


Figure 1. Cartoon representation of the automated experimental setup used for evaluating in vitro cell viability following ultrasound exposure.

medium containing 10% FBS and 50 μM 2-mercaptoethanol) supplemented with 200 nM PMA in a humidified incubator at 37 $^{\circ}\text{C}$ with 5% CO_2 (Forma Series II Water-Jacketed CO_2 Incubator, 184 L). Cells were then washed twice with DPBS (containing MgCl_2 and CaCl_2), and 2 mL of a fresh base medium was added and incubated for another 24 h prior to ND labeling. When used for confocal microscopy, the above protocol was followed by plating 2×10^5 THP-1 cells in a 35 mm glass-bottom Petri dish (P35G-1.5-14-C, Mattek, MA, USA).

2.4. ND Labeling of THP-1 Macrophages. To validate the internalization of NDs, we dispersed 10^8 to 10^{10} DiD-labeled NDs in 1 mL of a complete culture medium that was then added to each well containing 10^6 THP-1 cells. After 30 min of incubation at 37 $^{\circ}\text{C}$, cells were washed four times with 1 mL of 1 $\times$ DPBS to remove all free NDs. Cells were then imaged using an inverted phase-contrast microscope (CKX41, Olympus Inc., Tokyo, Japan) equipped with a CCD camera (DP73, Olympus Inc., Tokyo, Japan). A Cy5 filter cube was used to collect the DiD signal.

To localize NDs within cells, 2×10^9 Cy5-NDs were dispersed in 200 μL of a RPMI-1640 complete medium and added to 35 mm glass bottom Petri dishes containing 2×10^5 THP-1 cells (10,000 NDs/cell). After 30 min of incubation at 37 $^{\circ}\text{C}$, cells were washed four times with 500 μL 1 $\times$ DPBS to remove all free NDs and incubated with 1 $\mu\text{g}/\text{mL}$ Calcein-AM to stain the cytosol of live cells—Calcein-AM only fluoresces when it reacts with esterases in live cells and 10 $\mu\text{g}/\text{mL}$ Hoechst 33258 to stain nuclei. Thirty minutes later, cells were washed twice with 500 μL of 1 $\times$ DPBS, and the Petri dish was placed on the confocal microscope stage and kept at 37 $^{\circ}\text{C}$ during the entire image acquisition. Images were acquired as Z-stacks with 1 μm slice thickness using a Zeiss 880 inverted confocal microscope and a 63 $\times$ 1.40 NA oil immersion objective. Blue ($\text{Ex} = 405 \text{ nm}$, $\text{Em} = 455 \text{ nm}$) to image nuclei, green ($\text{Ex} = 488 \text{ nm}$, $\text{Em} = 553 \text{ nm}$) to image the cytosol of live cells, and red ($\text{Ex} = 633 \text{ nm}$, $\text{Em} = 698.5 \text{ nm}$) to image Cy5-labeled NDs were acquired and post-processed using the Fiji software.³⁶

2.5. Acoustic Droplet Vaporization of Internalized NDs In Vitro. THP-1 macrophages were prepared, incubated with NDs at 1000 NDs/cell for 30 min, washed four times to remove all free NDs, and trypsinized with 0.25% Trypsin–EDTA at 37 $^{\circ}\text{C}$ for 10 min, and the cell suspension was collected in 2 mL of a full medium and centrifuged at 500g for 5 min. Pelleted cells were resuspended in 3 mL of a full medium saturated with PFB and transferred to a 3 mL plastic transfer pipette bulb. The resulting cell concentration was 10^6 cells/mL. A pipette holder was made of 1% agarose gel containing 0.5% cornstarch molded in a 50 mL conical centrifuge tube around an empty 3 mL transfer pipette bulb that was removed after the agar gelled. The pipette filled with cells was placed in the holder and submerged in a 37 $^{\circ}\text{C}$ water tank for US imaging. After 5 min, cell suspensions were imaged using the 18L6 transducer (7 MHz central transmit frequency) in contrast pulsing sequence (CPS) mode of a Siemens Acuson Sequoia 512 scanner (Siemens Medical Solutions, Mountain View, CA). The CPS image was reconstructed from only

the nonlinear harmonic signals that were generated uniquely from MBs. Since the CPS image has a minimal signal, it is displayed side by side with the brightness mode (B-mode) image for orientation. Because real-time CPS imaging is not contaminated by the tissue signal, it has extreme sensitivity to MBs. The mechanical index (MI), a measure of transmitted US energy, is continually displayed on clinical scanners by FDA requirements to inform the sonographer of the energy used. Imaging the cell suspension began at very low MI to avoid ADV and was slowly increased until MBs were observed, indicating ADV of contained NDs.

2.6. Immobilization of ND-Labeled THP-1 Macrophages in the Hyaluronic Acid (HA) Hydrogel. Embedding of ND-labeled THP-1 macrophages in the hydrogel was performed to mimic the in vivo scenario where cells would be stationary and surrounded by tissues and to avoid direct exposure of US on a single-cell layer. The procedure was executed following a published protocol with modifications.³⁷ Briefly, differentiated THP-1 macrophages were labeled with 50 NDs/cell for 30 min at 37 $^{\circ}\text{C}$. The cells were then washed four times with PBS and incubated with 0.25% Trypsin–EDTA at 37 $^{\circ}\text{C}$ for 10 min. Next, the wells were scraped to recover all cells and trypsin was inactivated by adding four times the volume of the medium. The cell suspension was then centrifuged at 500g for 5 min, and pelleted cells were resuspended in 1 mL of the medium supplemented with 100 $\mu\text{g}/\text{mL}$ DNase and incubated for 10 min at room temperature. This step was added to digest the loose DNA that was released during the cell recovery from the plates because differentiated macrophages are highly adherent and require extra effort to detach, which can result in DNA release. Moreover, the digestion of DNA at this step is necessary for a fair assessment of cell death caused solely by ADV later on. The culture medium (19 mL) was added before centrifugation at 300g for 5 min. From this step forward, only the PFB-saturated medium was used. Cells were then resuspended in 150 μL of the medium and counted with a cellometer (Nexcelom Biosciences), and a mixture was created with 10^6 live cells/mL, 1.5–2% wt/vol methacrylate anhydride modified hyaluronic acid (HAMA), and 0.32% wt/vol lithium acylphosphonate (LAP). The mixture (250 μL) was transferred in a 24-well plate, and the HAMA polymer in solution was crosslinked by UV exposure at 254 nm for 1.5–2 min using a Spectrolinker XL-1000 (Spectronics Corporation). The same volume of medium was then added on top of the gel. The final percentage of HAMA and the duration of UV exposure were adjusted depending on the different batches of polymers to obtain a similar gel stiffness.

2.7. Acoustic Droplet Vaporization of Internalized NDs in Macrophages Immobilized in the HA Hydrogel. **2.7.1. Method 1: Manual Handling of the Transducer.** A circular cutout the size of a single-well plate was made in a 5 mm-thick foam to prevent the sound waves from reaching the other wells, the foam place on a thin plastic membrane, and the cavity filled with ultrasound gel. The well of interest containing hydrogel was placed over the foam and insonated from below using the 10L4 transducer. ADV pulses were delivered using the automated “Burst Mode” on the scanner that

delivers 1 pulse/s at 4 MHz central frequency and at an MI of 1.4 (transmission power 0.2%, low frequency). While we did not doubt that the higher frequency pulses delivered by the 18L6 transducer would trigger ADV, we used the 10L4 transducer to mimic the *in vivo* scenario when insonating deeper tissues. To confirm intracellular MB formation, the hydrogel was transferred to a 24 × 50 mm glass coverslip immediately after insonation, and bright-field confocal images were captured using an Olympus CKX41 microscope.

2.7.2. Method 2: Automated Handling of the Transducer. Immediately after gelation, ADV was performed using the 10L4 transducer on "Burst Mode" (transmission power 0.2%, low frequency, MI 1.4) as previously done but with the following modifications. To reduce experimental variations, the transducer was secured using a clamp connected to a custom-made water piston and fabricated using plastic syringes of different diameters (see the photograph in Figure S1). The device was secured on a ring stand, and the piston was connected to a peristaltic pump that allowed free and steady movement at a fixed rate of 60 mL/min. To perform ADV, the plates were placed on top of a foam holder with a central cutout the size of a single well, preventing US from traveling to surrounding wells. All of which was resting on a plastic wrap placed on top of a cardboard cutout that served as the physical support for the 24-well plates. The US transducer was placed underneath the plastic wrap, and US gel was applied on both sides (see Figure 1 for details). The peristaltic pump and "Burst Mode" were then activated simultaneously, enabling steady and homogeneous exposure to US in each well. Immediately after ADV, the presence of intracellular MB formation was confirmed using a Cytation 5 microscope (Agilent Biotek).

2.8. Cell Viability Assessment Following ADV. **2.8.1. Cell Viability Assessment Immediately after ADV.** To assess macrophage viability immediately after ADV, 200 μ L of PFB saturated 1× DPBS containing Calcein-AM (1 μ g/mL) was added on top of the hydrogel immediately following ADV (Method 1). After 30 min of incubation at 37 °C, to allow the label to reach the cells and interact with live cells with active esterases, the hydrogel was transferred to a 24 × 50 mm glass coverslip and fluorescence images were captured using the Olympus CKX41 microscope. Note that while Calcein-AM enters live, live and permeabilized, and dead cells, it only fluoresces when it enters live cells after it interacts with intracellular esterases. Three independent experiments were performed to validate the observations. Calcein-AM labeling was also performed on cells exposed to US but without prior incubation with NDs as well as on cells and ND-labeled cells without US exposure.

2.8.2. Cell Viability Assessment 3.5 h Post ADV. To assess macrophage viability after ADV (Method 2), 50 μ L of the WST-8 reagent was added to each well and incubated for 3 h at 37 °C. The supernatant (100 μ L) was then collected and transferred in a 96-well plate, making sure not to damage the hydrogel, and absorbance was read at 460 nm using a Synergy H1 microplate reader (Agilent Biotek). To further assess the viability, 150 μ L of the PFB-saturated medium containing 4 μ M of Calcein-AM and 8 μ M of Ethidium homodimer-1 from a live/dead viability/cytotoxicity kit was added to each well and incubated for 30 min at 37 °C (final respective concentrations of 1 and 2 μ M). The fluorescence signal from live and dead cells was then captured using a Cytation 5 microscope. For each condition, several pictures were acquired at different locations and different depths within the hydrogel. An average of 800 total cells was captured per condition, for each replicate.

All images were processed using the FIJI software³⁶ to filter out out-of-focus cells using background subtraction as well as circularity and size exclusion parameters (see Figure S2 for example image).

2.9. Assessment of ADV *In Vitro* at Various Times after Cell Labeling. Adherent THP-1 macrophages were incubated with 100 NDs/cell for 1 h, washed four times with DPBS, trypsinized, isolated, and embedded in the HA hydrogel as before using a 24-well plate with 10⁶ cells/well and kept at 37 °C. Using the 10L4 transducer, cells were imaged from below using the foam standoff and ADV was induced using the "Burst Mode" (Method 1). Wells were insonated either immediately after hydrogel crosslinking or at 1.5, 3, 8, or 12 h.

Immediately after US exposure, wells were imaged using a multi-mode reader (BioTek Cytation 5) to assess if MBs were generated. Each condition was repeated in three different wells.

2.10. Assessment of Phagocytosis Function Post ADV. Macrophages were loaded with PFB NDs by incubating them with 100 NDs/cell for 1 h, washed, and resuspended in a complete medium as before. A circular 1 mm-thick silicone grease ridge (Dow Corning High Vacuum Grease) 1 cm in diameter was made on a Petri dish pre-coated with poly-D-lysine to limit cell motion. The cell suspension (20 μ L) was placed in the formed well and incubated for 30 min to allow cells to adhere. Bright-field microscopy was performed with a Cytation 5 inverted imaging system using a 20× objective at 37 °C. The Petri dish was removed and insonated from below as before (Method 1), 20 μ L of pHrodo Deep Red *E. coli* BioParticles solution (1 mg/mL) was added to the well, the well was covered immediately with a glass coverslip, and the Petri dish was incubated for 2 h. Wells were reimaged with the Cytation 5 after incubation that allows for accurate repositioning as was done before labeling. Fluorescence images were analyzed using The Fiji software.³⁶

2.11. *In Vivo* Ultrasound Imaging and ADV of PFB ND-Labeled Macrophages. All animal experiments were approved by the University of Texas Southwestern Medical Center Institutional Animal Care and Use Committee (IACUC) and carried out in compliance with institutional guidelines. As a proof of concept, THP-1 cells were incubated with either 10,000 or 1000 NDs/cell for 30 min or 1 h, respectively, washed four times, trypsinized, and suspended in saline at 10⁷ cells/mL. Two 8 week old female nude rats were anesthetized with isoflurane (2% at 1.5 L/min using medical air) and placed on a heating pad, and the tail vein was cannulated. Body weight sodium nitroprusside (1 μ g/g) dissolved in 100 μ L of saline was infused intravenously. Five minutes later, 5 × 10⁶ cells suspended in 500 μ L of saline were infused intravenously. Rat-1 was infused with the cells incubated with 10,000 NDs/cell, and rat-2 with cells was incubated with 1000 NDs/cell. Rat-1 was scanned at 45 min after infusion and rat-2 at 1.5, 4.5, and 8 h after infusion. CPS imaging of the liver and spleen with the 18L6 at 0.06 MI was done to determine whether MBs had formed by spontaneous vaporization. While imaging in CPS at low MI with the 10L4 transducer, the focal zone was positioned at the far-field of the liver and ADV pulses delivered using the built-in "Burst Mode". After ADV, the liver was imaged with CPS using a dual-screen display at low MI intermittently until MBs were no longer visible. Rat-1 was euthanized at 45 min, and the spleen was removed intact and scanned in a Petri dish before and after ADV to confirm that MB formation occurred in only the region insonated.

3. RESULTS AND DISCUSSION

3.1. ND Formulation and Characterization. PFB NDs were made with varying ratios of DSPC to DSTAP to produce 0–10, 20, and 30% of DSTAP relative to total lipids while keeping the DSPE-PEG 2K at 10%. The purpose of exploring different DSTAP contents was to formulate NDs with a positive surface charge to facilitate phagocytosis aided by the electrostatic interaction with the negatively charged cell membrane. The different ND formulations only exhibited small variations in size and concentration (Figure S3). This was expected because DSTAP is a saturated lipid that does not impact the strong attractive intermolecular forces and molecular packing of phospholipids' saturated alkyl chains of equivalent lengths. However, increasing the DSTAP content markedly increased the zeta potential (Table 1). As the 30% DSTAP formulation produced the highest positive zeta potential and the highest colloidal stability and therefore the highest potential to interact with the negatively charged cell membrane, it was selected and used for all remaining cell labeling experiments. The gas volume generated after heating 50 μ L of a 9% v/v PFB emulsion was ~700 μ L, which is in agreement with the theoretical 690 μ L expected (Figure S4a)

Table 1. Size, Zeta Potential, and Count Distribution of Different Formulations Measured by the Nanoparticle Tracking Analyzer^a

formulation type	mean diameter	zeta potential (mV)	concentration (#/mL)
DSPC:DSPE-2K (90:10)	286 ± 50	−5.3 ± 4.9	$5.9 \times 10^{11} \pm 4.6 \times 10^{11}$
DSPC:DSPE-2K:DSTAP (80:10:10)	319 ± 15	+4.5 ± 7.5	$6.5 \times 10^{11} \pm 4.7 \times 10^{11}$
DSPC:DSPE-2K:DSTAP (70:10:20)	346 ± 28	+14.8 ± 10.6	$1.0 \times 10^{12} \pm 5.4 \times 10^{11}$
DSPC:DSPE-2K:DSTAP (60:10:30)	305 ± 25	+38.0 ± 6.1	$7.1 \times 10^{11} \pm 3.5 \times 10^{11}$

^aValues are given as mean ± SD, *n* = 3 for each group.

for an encapsulation efficiency of nearly 100%. The presence of PFB in NDs was also confirmed by gas chromatography/mass spectrometry (GC–MS) (Figure S4b,c).

DLS confirmed ND stability in cell culture media as no significant changes in intensity, intensity-weighted size distribution, derived count rate, and correlation curves were detected over the 1 h observation period compared to the control sample in milliQ water (Figure 2). The absence of ND loss due to spontaneous vaporization caused by thermal instability was confirmed by the stable signal intensity in the intensity-weighted size distribution and the absence of signal above 1 μm (Figure 2a). In addition, the absence of change in both correlation curves' Y-intercept and exponential decay lifetime also suggests that NDs did not phase change to create MBs and liposomes (Figure 2b).³⁰ While protein corona formation has been reported to induce nanoparticle agglomeration, potentially leading to a significant size increase and unusable results,³² our NDs only exhibit a small change in size or derived count rate (Figure 2c), both in agreement with ND stability.

3.2. Validation of ND Internalization by THP-1 Macrophages. We compared the internalization efficiency of different ND formulations using human THP-1 derived macrophages. Due to their nanoscale size, NDs are not visible by bright-field microscopy, which precludes the assessment of their integrity after internalization. We hypothesized that the ND structure will not be disrupted during phagocytosis as they are constituted of the same lipids as MBs, and MBs remained stable post internalization in neural progenitor cells¹⁹ and stem cells.²⁵ We exposed the adherent cells in culture to DiD-labeled NDs formulated with 30% DSTAP at 100, 1000, or 10,000 NDs/cell for 30 min. As shown in the fluorescence

microscopy images in Figure 3a, NDs were phagocytosed by macrophages at all ND/cell ratios. We confirmed their

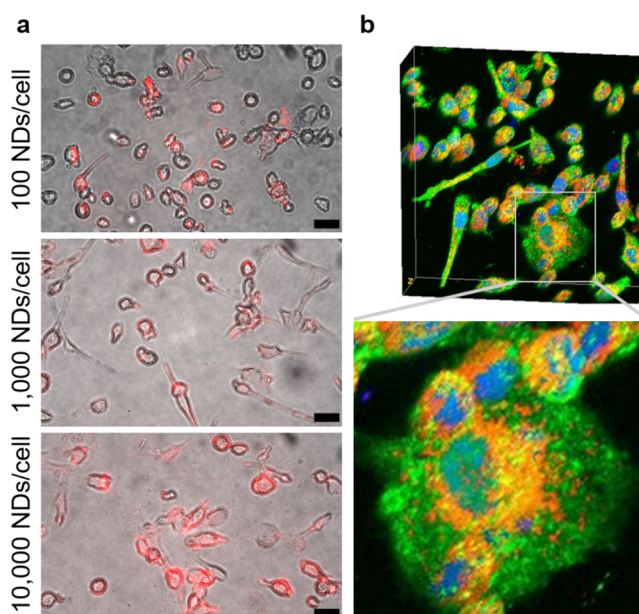


Figure 3. In vitro optical imaging of ND internalization in THP-1 macrophages. (a) Overlay of fluorescence and bright-field images showing strong association of DiD-labeled NDs (scale bars = 50 μm) with cells. (b) Representative confocal microscopy image (Z-stack) of macrophages after incubation with 30% DSTAP-Cy5-NDs at 1000 NDs/cell: NDs (red), Hoechst (blue), and Calcein (green) show internalization of the discrete endosomal DiD signal.

intracellular location with confocal microscopy using Cy5 labeled PEG rather than the lipid soluble DiD dye to ensure that the dye does not transfer or other lipid surfaces. After incubation with 1000 NDs/cell, cells were counterstained with Hoechst and Calcein-AM to assess cell viability. As presented in Figure 2b, NDs were in the cytoplasm in the perinuclear region, which is typical of phagocytosed particles. Moreover, the Calcein-AM staining showed that cells were still alive after the process. It is worth noting that no MBs were observed during the labeling, staining, and imaging process, indicating that no spontaneous vaporization occurred. In addition, cell counting using a live/dead stain after ND labeling showed 90% viability, which was similar to unlabeled cells (Figure S5).

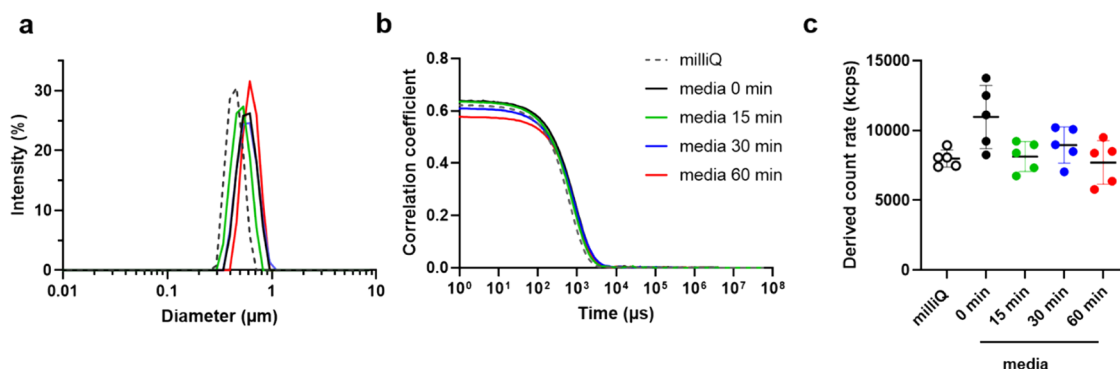


Figure 2. Stability of NDs in media assessed by DLS. Representative (a) intensity-weighted size distributions, (b) correlation curves, and (c) derived count rates (*n* = 5) as a function of incubation time in media at 37 °C.

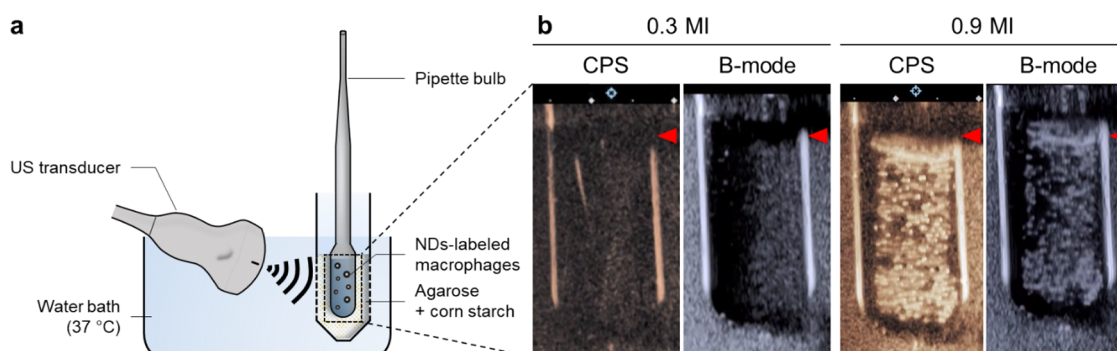


Figure 4. In vitro ultrasound detection of acoustic vaporization of internalized NDs. (a) Schematic representation of the experimental setup used. ADV and image acquisition were performed using a clinical US system. (b) Representative images acquired at 0.3 and 0.9 MI. The echoes observed within the bulb on B-mode at 0.3 MI are due to partial volume artifacts that were visible prior to increasing the MI (see Video S1). Note the dramatic increase in the signal throughout the sample, particularly at the air–liquid interface (red arrowhead) at 0.9 MI.

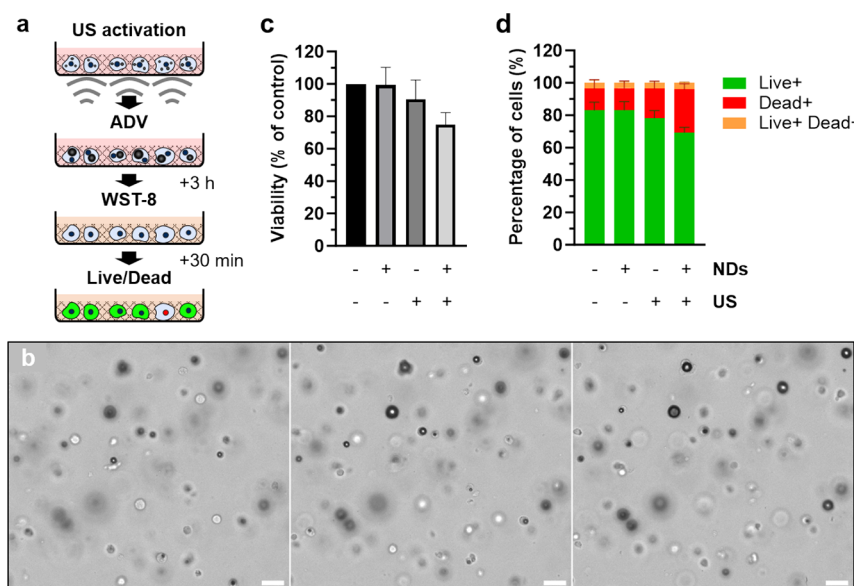


Figure 5. Viability assessment of ND-labeled THP-1 macrophages embedded in the hydrogel following ADV. Macrophages were exposed to 50 NDs/cell for 30 min before ADV. (a) Schematic representation of the experimental workflow. (b) Representative bright-field images of MBs in macrophages following ADV. Each panel was acquired at different heights but identical locations within the hydrogel. MBs appear as bright spheres surrounded by a black edge. Scale bars = 50 μ m. (c) WST-8 assay (results are normalized to control cells). (d) Live/dead assay microscopy quantification. Results expressed as mean $\pm$ SEM; $n = 4$.

3.3. Ultrasound Detection of Acoustic Vaporization of Internalized NDs in Suspension. To determine the MI at which acoustic droplet vaporization (ADV) of intracellular NDs occurs, ND-labeled THP-1 macrophages were suspended in a 3 mL plastic transfer pipette at 10^6 cells/mL and imaged at 37 $^{\circ}$ C using a dual display of CPS images and their corresponding B-mode. Using the 18L6 transducer transmitting at 7 MHz central transmit frequency, no MBs were present when imaged at a low MI of 0.06, indicating that no spontaneous vaporization had occurred during cell and sample preparation and heating to 37 $^{\circ}$ C. As the MI was increased, MBs began to form at an MI of 0.6 at the air–liquid interface and at the focal zone where energy deposition is greatest and then throughout the sample at an MI of 0.9 (Figure 4 and Video S1). In a previous study, we observed that free PFB NDs in suspension formulated without DSTAP vaporized at an MI greater than 0.4.³¹ Note that since we cannot control the number of internalized NDs per macrophage, which impacts the ADV threshold,³⁸ this experiment suggests that internalized

NDs require more energy to vaporize as compared to free NDs in suspension, albeit at less than half the maximum MI allowed by the FDA for clinical US systems.

3.4. Cell Viability of Cells Containing Vaporized NDs Immediately after ADV. Reports in the literature suggest that ADV at MI > 1.9 can impact both cell viability and biological function after the phase transition.^{27,28} It was also reported that as the zeta potential of cationic particles increases, it can destabilize endosomal membranes, increase the production of reactive oxygen species, and induce apoptosis.³⁹ Since our approach utilizes both cationic particles and ADV, we aimed to confirm that cell viability and biological functions were preserved after vaporization of internalized PFB NDs, both mandatory for in vivo applications. Thanks to the excellent sensitivity of US at detecting MBs, we hypothesized that lowering the number of NDs per cell would not negatively impact US detection in vivo as we can in theory detect a single MB post ADV. THP-1 cells were exposed to 50 NDs/cell for 30 min, washed, trypsinized, and suspended in an optically

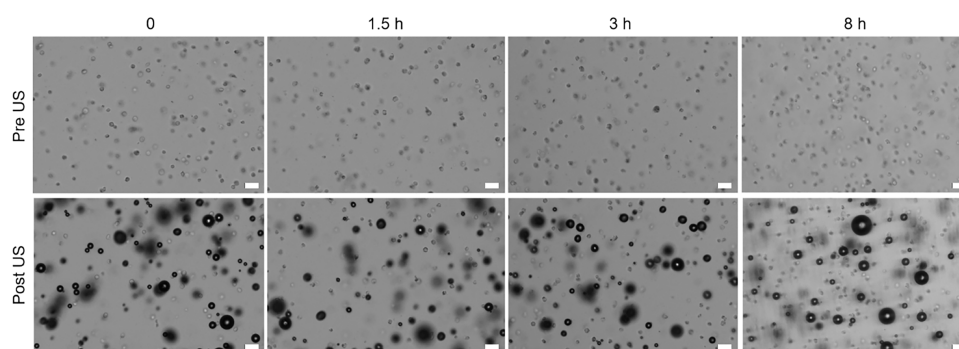


Figure 6. Representative microscopic images show MBs after ADV in THP-1 macrophages from shortly after embedding in the hydrogel and up to 8 h after ND labeling (scale bars = 50 μ m).

transparent HA hydrogel ($45 \times 25 \times 2$ mm) in which cells remain alive and can divide and grow.³⁷ No MBs were observed when imaged by US at low MI, indicating ND stability during culture and manipulation. Manually performed ADV (Method 1) with the 10L4 transducer induced MB formation (Figure S6a,b). Calcein-AM, a non-fluorescent and membrane permeable compound, was then added to the top surface of the porous hydrogel, to assess cell viability immediately after ADV. Microscopy was done 30 min after ADV to assess whether cells containing MBs were alive. As apoptotic and dead cells with compromised cell membranes lack esterase activity and do not retain the Calcein-AM label, the presence of green fluorescence after ADV indicates that cells are alive with intact membranes as they enzymatically converted Calcein-AM into Calcein post insonation that was retained within the cell.

Interestingly, despite the many NDs phagocytosed, we could only detect one intracellular MB after activation and it had a diameter of ~ 10 μ m. We believe that once MBs form, they either coalesce and/or their gas core inflates because of the co-existing non-vaporized NDs.³¹ No difference was observed between US-treated and non-isonated cells, with or without incubation with NDs (Figure S6c). We believe that the relatively low number of macrophages with vaporized NDs is most likely due to the short-lived newly formed MBs rather than labeling efficiency or differing vaporization thresholds.

3.5. Cell Viability Assessment up to 3.5 h Post ADV.

Using the automated handling of the transducer (Method 2), cell viability was assessed post ADV using two complementary techniques at longer time periods (Figure 5a,b). First, the cells were incubated with WST-8, a water-soluble reagent that produces a color change when metabolized by live cells. As presented in Figure 5c, while the incubation of cells with NDs did not decrease cell viability, exposure to US alone reduced viability by 10% and ADV caused an additional 15% reduction. Those trends were confirmed with the live/dead assay presented in Figure 5d. This assay revealed a reduction of 5% in the number of live cells exposed to US compared to control cells and cells treated with NDs. The combination of NDs and US (i.e., intracellular ADV) slightly decreased cell viability by 14%. It is important to note that the live/dead assay was purposefully performed several hours after ADV to allow the cell membranes to recover from potential sonoporation resulting from US exposure. Otherwise, the ethidium homodimer-1 used to label dead cells could potentially cross the membrane and stain the nucleus, creating false positives. Moreover, when imaging the cells 3.5 h post ADV to quantify the live/dead ratio, no MBs were ever

observed inside the cells. This result confirms that there will not be any further toxicity resulting from MB inflation after this time point.

Taken together, these results show that US slightly decreases cell viability and that the intracellular formation of MBs by ADV further decreases viability. However, this loss of viability remains minimal as the cells maintain 75% of their metabolic activity post-ADV and more than 83% of the cells remain alive compared to control cells. While this *in vitro* experiment shows successful intracellular ADV in viable cells, it also does not reflect the cell viability *in vivo* where said cells are surrounded by soft tissues, which will lower the negative impact on viability.

3.6. Delayed ADV of Internalized NDs. Immediately after internalization, PFB NDs did not spontaneously vaporize until insonated at sufficient energy (Figure 4). However, the highly volatile PFB could escape the NDs over time when cells are exposed to air at atmospheric pressure both *in vitro* and *in vivo*, particularly when cells in the circulation pass through lung capillaries. We then defined the time window for ADV after internalization using cells exposed to 100 NDs/cell for 1 h and suspended in the HA hydrogel shortly after internalization. ADV was performed shortly after or after 1.5, 3, 8, or 12 h later and MB formation confirmed by microscopy. As before, cells that were not exposed to US showed no MB formation up to 8 h after internalization. A large number of cells containing NDs became visible after ADV up to 8 h post internalization (Figure 6). ADV was not visible at 12 h (Figure S7), indicating that sufficient NDs remain stable and available for ADV for at least 8 h, which allows for an 8 h time window for cells to home to their target after systemic administration, prior to activation.

3.7. Phagocytosis Function after ADV. As a functional assessment of cell survival following ADV, we tested the ability of macrophages to maintain their phagocytic function after ADV, an important primary function of macrophages as professional antigen presenting cells. To do so, we used pH-sensitive bioparticles that fluoresce red only in the acidic environment expected in both late endosomes and lysosomes, indicating successful phagocytosis.^{40,41} For this experiment, we used PFB ND-loaded macrophages that were plated on a poly-D-lysine-coated Petri dish to limit cell motion during labeling. A circular ridge of silicon grease formed a well to keep the cells bathed in growth media. While we favor the use of hydrogel to better predict ADV performance in tissues, we used adherent cells for this experiment as the hydrogel limits access of the pH-sensitive beads to the embedded cells. The beads that were added to the formed well after ADV, sealed using a glass

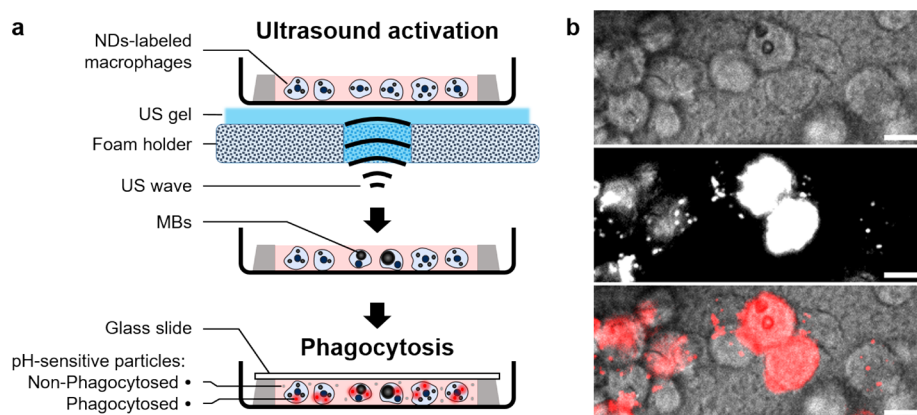


Figure 7. Phagocytosis ability of ND-labeled THP-1 after ADV. (a) Schematic representation of the experimental setup. (b) Representative bright-field, fluorescence, and overlay images of THP-1 macrophages after ADV that now contained MBs and allowed to phagocytose the pH-sensitive bioparticles. MBs appear as black-rimmed circles. (Scale bars = 25 μ m).

coverslip, and incubated for 1 h were imaged with the Cytation 5 microscope (Figure 7a). Note that cells containing 4–20 μ m MB became fluorescent (Figure 7b and Figure S8), indicating that cells were able to phagocytose the beads after ADV. Phagocytosis was also validated in the absence of NDs as well as using ND-loaded cells without ADV as controls (Figure S8). Note that not all the cells showed red fluorescence from the beads in the ADV or control groups. One additional observation and limitation of our work is the fact that fluorescently labeled cells without MBs may or may not have hosted MBs as MBs may disappear during the 1 h incubation period after ADV.

3.8. In Vivo Ultrasound Imaging of Intracellular PFB NDs Vaporization. We showed above that macrophages remain alive after phagocytosing a large number of NDs and after ADV. Further, macrophages with internalized ND that underwent ADV were able to phagocytose bioparticles in vitro. Before developing this approach as a cell-based theranostic, we needed to determine whether vaporization occurs in vivo. To this end, THP-1 cells were exposed to 10,000 NDs/cell (Figure 8) or 1000 NDs/cell (Figure 9) for 30 min, washed to

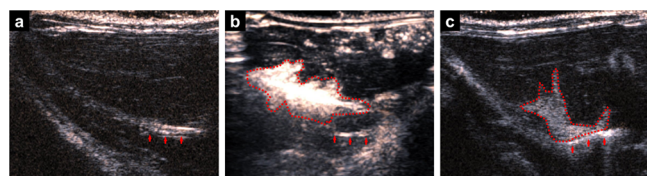


Figure 8. In vivo detection of ND-labeled macrophages after phase change using a clinical scanner. Representative image of the liver: (a) acquired with the 18L6 transducer at low MI prior to ADV showing no MBs; (b) acquired during the "Burst Mode" using the 10L4 showing intense CPS signal generation; (c) acquired 30 min post ADV with the 18L6 transducer at low MI showing persistence of the MB signal. The dotted line region of interest highlights the MB local signal enhancement, and red arrowheads indicate the diaphragm. The 2D field of views is 20 mm deep $\times$ 28 mm wide.

remove free NDs, trypsinized, and suspended in PBS at 10^7 cells/mL. Before injection, nude rats were pre-treated with an intravenous dose of 1 μ g/g body weight of sodium nitropruside to minimize cell trapping in the lungs.^{42,43} Five minutes later, 5×10^6 ND-loaded macrophages were infused through the tail vein. Since cells given systemically first accumulate in

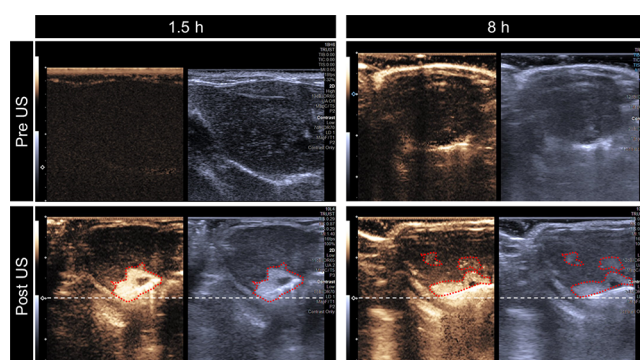


Figure 9. In vivo detection of ND-labeled macrophages after ADV using a clinical scanner 1.5 and 8 h post intravenous infusion. Images of the liver acquired with the 18H6 (1.5 h) or 10L4 (8 h) at low MI prior to ADV show no MBs (top row), and those acquired with the 10L4 during the "Burst Mode" (bottom row) show intense MB signal formation on both CPS (orange) and B mode (gray-scale), particularly at the focal zone (dashed white line). Dimensions of the field of views are 20 mm deep $\times$ 20 mm wide at 1.5 h and 40 mm deep $\times$ 35 mm wide at 8 h.

the lung, liver, and spleen, the liver of one rat was imaged 45 min after cell administration at high frequency and low MI to ensure that no ADV occurs. No MBs were detected in the liver, indicating that no spontaneous vaporization occurred during cell preparation and administration (Figure 8a). The rat's liver was then imaged at low MI using the 10L4, which was used as an image guide to induce ADV in the far-field of the liver. ADV was performed using the built-in "Burst Mode" that is intended to burst MBs in the field of view to enable blood flow measurements. This mode delivers energy at slightly greater than 1 MI, which is still below the FDA limit of 1.9 MI (Figure 8b). The liver was imaged again with the high frequency transducer at low MI to minimize MB destruction (Figure 8c). The MB signal faded and became invisible at 30 min. The rat was killed, and the spleen was removed to image ex vivo. ADV was observed exactly where the pulses were delivered (Figure S9). Since we now knew that ADV occurs in vivo after NDs accumulated in the liver and spleen, we loaded macrophages with less NDs and studied the length of time that NDs will be available for vaporization after infusion. In another rat, 5×10^6 THP-1 cells incubated with 1000 NDs/cell were injected intravenously, and ADV was performed over the liver at 1.5,

4.5, or 8 h later (Figure 9 and Video S2). Similar to the in vitro data, MB formation was observed up to 8 h but not at 12 h likely because of PFB elimination. When the same number of NDs (1000 NDs/cell $\times$ 5×10^6 cells) was infused intravenously as free NDs in suspension, MB formation in the liver occurred at 1.5 but not at 4 h. No MBs were observed in blood at either time point before or after ADV pulses. Since PFC NDs accumulate in the reticuloendothelial cells of the liver,⁴⁴ these observations indicate that internalized NDs persist longer than circulating NDs in vivo (Figure S10).

4. CONCLUSIONS

With the long-term objective of developing cell-based theranostics that can be turned on and detected by a clinical US scanner, we showed that stable positively charged NDs containing low boiling point perfluorocarbon were internalized by macrophages without affecting cell survival. Internalized NDs could be vaporized into MBs after being exposed to US pulses at less than half the FDA power limits. This vaporization did not affect cell survival or phagocytic function. ND vaporization was successfully achieved for at least 8 h after internalization both in vitro and in vivo, leaving 8 h for cells to home to their target in future therapeutic strategies. In this proof-of-concept work, we used macrophages as the cells to acoustically label as they are known to home to tumors and injured tissues. Future efforts will focus on whether a sufficient number of cells can home to their target by 8 h to express the desired biologic effect when bioactive molecules/genes carried on the ND surface are released into the cytosol upon activation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.3c11976>.

Quantification of the PFB content by direct visualization and by GC–MS, internalization of DiD NDs formulated with 0–10% DSTAP, macrophage viability after ND internalization with or without US exposure, macrophage phagocytosis after ND internalization with or without US exposure, ex vivo ND activation in the spleen, and in vivo ultrasound imaging of free NDs (PDF)

In vitro ultrasound detection of acoustic vaporization of internalized NDs (AVI)

Ultrasound imaging and tracking of vaporized intracellular NDs after macrophage accumulation in the liver (AVI)

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Author Contributions

[#]L.C. and J.S. contributed equally. The manuscript was written through contributions by all authors. C.d.G.L. and R.F.M. contributed to the conceptualization and design of the study. L.C., J.S., A.W., N.H., S.G., and R.F.M. performed the experiments. C.E. assisted in conducting the animal experiments. L.C., J.L., J.S., C.d.G.L., and R.F.M. analyzed the data. J.S., L.C., and C.d.G.L. were responsible for the figure production. C.d.G.L. prepared the manuscript. L.C., J.S., J.L., and R.F.M. revised the manuscript. C.d.G.L. supervised the whole study. All authors have given approval to the final version of the manuscript.

Notes

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